# SARAH LIBANORE

Postdoctoral Fellow

**Keywords:**
Observational Cosmology
Large-scale structure of the Universe

+39 342 8065889 | sl2412@cam.ac.uk // sarah.libanore@gmail.com
0000-0002-2284-9190 | github.com/slibanore | https://slibanore.github.io

## RESEARCH INTERESTS

Research Associate in astrophysics and cosmology, with a focus on cross-correlations. Currently working on CMB data and line-intensity mapping surveys, with research interests in primordial non-Gaussianity and the Epoch of Reionization, alongside previous experience in gravitational wave surveys. Passionate about teaching, student mentoring, collaborative research, and outreach.

## EDUCATION

**University of Padova**
**09.2018 - 05.2022**
**Ph.D. in Physics**
Advisors: Prof. Michele Liguori, Prof. Alvise Raccanelli
Thesis: *The importance of clustering analysis in future GW surveys*

**University of Padova**
**09.2016 - 09.2018**
**Master degree in Astronomy (110/110 cum laude)**
Advisors: Prof. Michele Liguori, Prof. Nicola Bartolo
Thesis: *Gravitational Wave detections: implications for Cosmology*

**University of Padova**
**09.2013 - 09.2016**
**Bachelor degree in Astronomy (110/110 cum laude)**
Advisor: Prof. Sabino Matarrese

## PROFESSIONAL EXPERIENCE

**DAMTP, U.Cambridge**
**10.2026 - Ongoing**
**Research Associate in Cosmology and Gravitation**
Advisor: Prof. Paul Shellard, James Fergusson

**Ben Gurion University**
**01.2023 - 09.2026**
**Azrieli International Postdoctoral Fellow**
Advisor: Prof. Ely D. Kovetz
Research Experience: line intensity mapping, simulation and forecast of 21-cm and star forming lines in the context of current surveys

**University of Padova**
**03.2022 - 12.2022**
**Assegno di Ricerca Pre-Dottorato**
Advisor: Prof. Michele Liguori

## GRANTS & AWARDS

- **BGU/Philadelphia academic bridge grant** (2025, co-PI with Profs. A. Lidz, E. Kovetz)
- **Balzan award**, held by Center for Cosmological Studies (2023, research visiting to JHU)
- **Azrieli International Postdoctoral Fellowship program** (2022)
- **Fondazione Aldo Gini scholarship** (2022)
- **PhD Sandwich Scholarship**, held by Council for Higher Education in Israel (2022)
- **Mille e una Lode**, held by University of Padova (2016)

## REFERENCES

- **Ely D. Kovetz**
  Ben Gurion University
  kovetz@bgu.ac.il
- **Julian B. Muñoz**
  University of Texas, Austin
  julianbmunoz@utexas.edu
- **Michele Liguori**
  Università degli studi di Padova
  michele.liguori@unipd.it
- **Alvise Raccanelli**
  Università degli studi di Padova
  alvise.raccanelli.1@unipd.it

## TEACHING & MENTORING

- **Tutorial** on "21cmFirstCLASS" (Santander, Spain, February 2025)
- **Assistant Lecturer** for "Cosmic Tensions" Ph.D. course (GGI school, Italy, March 2023)
- **Mentor of Ph.D. and Master students in their research projects**

**CURRENT**
H. Lazare (PhD, joint 21-cm+UVLF constraints)
K. Finish (PhD, fnl effect on 21-cm at EoR)
T. Kadosh (MsC, island model of EoR)

**PAST (5 papers)**
T. Adi (PhD, primordial magnetic fields and LIM)
J. Flitter (PhD, 21-cm during the EoR)
G. Shmueli (MsC, neutrino masses and LIM)
M. Peron (MsC, bias of GW mergers)

## COLLABORATIONS & MEMBERSHIPS

- Square Kilometer Array Observatory (SKAO)
- International Astronomical Union (IAU)
- European Consortium for Astroparticle Theory (EUCAPT)
- ULTRASAT
- COMAP

## OUTREACH & DISSEMINATION

**PLaNCK! magazine**
**2016 - ongoing**

**Editorial Board of science magazine for children**

Article writing and editing
Collaboration in public relations and social media management

**APS Accatagliato**
**2016 - ongoing**

**Outreach activities tailored to diverse audiences**

Planning, design, and implementation of workshops, festivals, and events
Science trainer & educator in schools, workshops, professional internships
Back-office support activities

**Professione... Scienza**
**2019-2021**

**Educational Game**

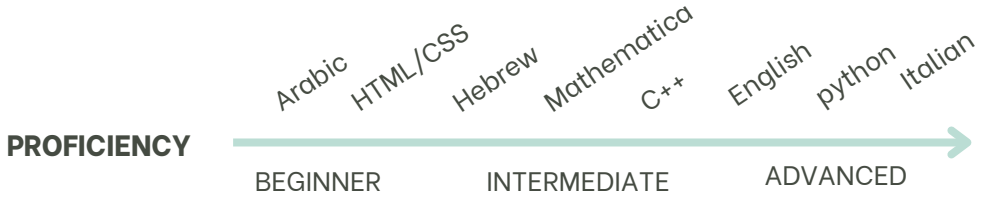
I collaborated in the development of educational game cards designed to engage young people with the work and role of scientists in scientific and societal progress. The cards were created with a focus on diversity and inclusion across gender, ethnicity, disability.

**Penne di Scienza**
**2024 - ongoing**

**Correspondence with school classes**

Engaged with teenagers via letters and online meetings to foster science communication and provide role modeling

## LANGUAGES



## TALKS & SEMINARS

- Israel–China Workshop on 21 cm Cosmology (online, October 2026)
- Cosmological Observables (Cambridge, September 2026)
- New Frontiers in Cosmology (A Coruna, August 2026)
- LIM26 (Pasadena, July 2026)
- Astrolunch (Hebrew University, June 2026)
- APG Seminar (SISSA,Trieste, May 2026)
- Astrophysics Seminar (Technion, May 2026)
- ARCO Seminar (Open University, January 2026)
- Dark Sector Meeting (Jet Propulsion Laboratory, November 2025)
- Cosmology Journal Club (Caltech, November 2025)
- Cosmology Group Meeting (University of Harvard, November 2025)
- Annual meeting of the Israeli Physical Society (Technion, July 2025)
- LIM 2025 (Annecy, June 2025)
- Cosmic Frontier Center Discussions (Austin, May 2025)
- SUBLIME meeting (Tokyo, March 2025)
- Astrophysics Seminar (Santander, February 2025)
- 21 cm Cosmology Workshop (Hangzhou, July 2024)
- LIM 2024 (Urbana Champain, May 2024)
- Astrophysics Seminar (Weizmann Institute, April 2024)
- Cosmology in the Alps (Les Diablerets, March 2024)
- Astrophysics Seminar (Bar Ilan University, February 2024)
- Astrophysics Seminar (Scuola Normale Superiore di Pisa, November 2023)
- HI as a Cosmological probe (Nazareth, May 2023)
- Present and Future of LIM (Garching, April 2023)
- Annual meeting of the Israeli Physical Society (Tel Aviv, April 2023)
- Cosmology Journal Club (Padova, November 2022)
- Asiago Cosmology Group Retreat (Asiago, September 2022)
- ICTP Cosmology Summer School (Trieste, July 2022)
- Astrophysics Seminar (Ben Gurion University, March 2022)
- Gravitational Wave day @ Unipd (Padova, December 2021)
- AstroParticle Symposium 2021 (Saclay, November 2021)
- CosmoClassic (online, September 2021)
- SKAstle (Tagliolo Monferrato, September 2021)
- First EuCAPT Annual Symposium (online, May 2021)
- IberiCOS 2021 (online, March 2021)
- University of the Western Cape Astro webinar (online, January 2021)
- CosmoMeetings at Universidade de São Paulo (online, July 2020)
- Cosmology Journal Club (Padova, June 2020)

## LIST OF PUBLICATIONS

FIND MY WORK ON INSPIRE HEP

### UNDER REVIEW

- N. Bellomo, M. Bosi, **S. Libanore**, et al. **(2026)**
  **arXiv:2609.09083 (submitted to JCAP)** - *Breaking binary formation mechanism degeneracies with gravitational wave clustering*
- H. Lazare, **S. Libanore**, E. Vanzan, J.B. Muñoz, E.D. Kovetz **(2026)**
  **arXiv:2607.20651 (submitted to Phys. Rev. D)** - *When galaxies burst II. Implications of enhanced burstiness for the 21-cm Cosmic Dawn signal*
- C. Scarpel, N. Bellomo, **S. Libanore**, M. Liguori, A. Raccanelli **(2026)**
  **arXiv:2607.12049 (submitted to JCAP)** - *Mitigating galaxy systematics with gravitational wave clustering*
- E.D. Kovetz, H. Lazare, **S. Libanore**, J.B. Muñoz, E. Vanzan **(2026)**
  **arXiv:2605.13967 (submitted to APJ Lett.)** - *When galaxies burst: enhanced shot-noise for line-intensity mapping in the JWST era*
- E. Thélie, **S. Libanore**, Y. Sklasnky, J.B. Muñoz, E.D. Kovetz **(2026)**
  **arXiv:2602.12277 (submitted to Phys. Rev. Lett.)** - *Reionization Bubbles from Real-Space Cross Correlations of Line Intensity Maps*

### PUBLISHED

18. **S. Libanore**, E.D. Kovetz, J.B. Muñoz, Y. Sklasnky, E. Thélie **(2026)**
    **arXiv:2509.08886v2 (accepted on Phys. Rev. D.)** - *A New Boundary Condition on Reionization*
17. M. Bosi, **S. Libanore**, N. Bellomo, et al. **(2026)**
    **(accepted for publication)** - *Chapter in Advancing Astrophysics with the SKA – II: Using SKAO to understand the clustering of gravitational wave sources*
16. D. Sarkar et al. (including **S. Libanore**) **(2026)**
    **(accepted for publication)** - *Chapter in Advancing Astrophysics with the SKA – II: Cosmology with Multi-Wavelength Line Intensity Mapping Synergies in the SKAO Era*
15. S. Majumdar et al. (including **S. Libanore**) **(2026)**
    **(accepted for publication)** - *Chapter in Advancing Astrophysics with the SKA – II: Cosmology with Intensity Mapping via statistics beyond the power spectrum in the SKAO Era*
14. **S. Libanore**, J.B. Muñoz, E.D. Kovetz **(2025)**
    **Phys. Rev. D 112, 083552 (2025), arXiv 2507.15922** - *Effective model for line intensity mapping: Auto- and cross-power spectra in the cosmic dawn and reionization*
13. **S. Libanore**, S. Ghosh, E.D. Kovetz, K.B. Body, A. Raccanelli **(2025)**
    **Phys. Rev. D 112, 063502 (2025), arXiv:2504.15348** - *Joint 21-cm and CMB Forecasts for Constraining Self-Interacting Massive Neutrinos*
12. G. Shmueli, **S. Libanore**, E.D. Kovetz **(2024)**
    **Phys. Rev. D 111 063512 (2025), arXiv:2412.04071** - *Towards a multi-tracer neutrino mass measurement with line-intensity mapping*
11. J. Flitter, **S. Libanore**, E.D. Kovetz **(2024)**
    **Phys.Rev.D 112 023537 (2025), arXiv:2412.04071** - *More careful treatment of matter density fluctuations in 21-cm simulations*
10. E. Vanzan, **S. Libanore**, L. Valbulsa Dall'Armi, N. Bellomo, A. Raccanelli **(2024)**
    **JCAP 10 014 (2024), arXiv:2405.13871** - *Gravitational wave background from primordial black holes in globular clusters*
9. **S. Libanore**, E.D. Kovetz **(2024)**
    **Phys. Rev. D 111 063521 (2024), arXiv:2404.01500** - *Upcoming searches for decaying dark matter with ULTRASAT ultraviolet maps*

8. **S. Libanore**, E.D. Kovetz **(2024)**
**A&A 687, A133 (2024), arXiv:2401.12285** - *Constraining z<2 ultraviolet emission with the upcoming ULTRASAT satellite*

7. **S. Libanore**, J. Flitter, E.D. Kovetz, Z. Li, A. Dekel **(2023)**
**MNRAS 532 1 (2024), arXiv:2310.03021** - *Effects of feedback-free starburst galaxies on the 21-cm signal and reionization history*

6. **S. Libanore**, M. Liguori, A. Raccanelli **(2023)**
**JCAP 08 055 (2023), arXiv:2306.03087** - *Signatures of primordial black holes in gravitational wave clustering*

5. M. Peron, A. Ravenni, **S. Libanore**, M. Liguori, M.C. Artale **(2023)**
**MNRAS 530 1 (2024), arXiv:2305.18003** - *Clustering of binary black hole mergers: a detailed analysis of the EAGLE+MOBSE simulation*

4. T. Adi, **S. Libanore**, H.A. Cruz, E.D. Kovetz **(2023)**
**JCAP 09 035 (2023), arXiv:2305.06440** - *Constraining Primordial Magnetic Fields with Line-Intensity Mapping*

3. **S. Libanore**, C. Unal, D. Sarkar, E.D. Kovetz **(2022)**
**Phys. Rev. D 106 123512 (2022), arXiv:2208.01658** - *Unveiling cosmological information on small scales with line intensity mapping*

2. **S. Libanore**, M.C. Artale, M. Liguori, M. Mapelli et al. **(2021)**
**JCAP 02 003 (2022), arXiv:2109.10857** - *Clustering of Gravitational Wave and Supernovae events: a multitracer analysis in Luminosity Distance Space*

1. **S. Libanore**, M.C. Artale, M. Liguori, M. Mapelli et al. **(2020)**
**JCAP 02 035 (2021), arXiv:2007.06905** - *Gravitational Wave mergers as tracers of Large Scale Structures*